

Supplementary material for: Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

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This document collects the supplementary figures (Figs. S1–S13), the specimen–thermal-history linkage (Table S1), the itemized bill of materials (Table S2), and the design-file inventory (Table S3) that support the main text. File paths are relative to the root of the openly available data and design repository (<https://github.com/vertical-cloud-lab/custom-induction-furnace>); an archival snapshot is deposited at Zenodo, [10.5281/zenodo.20878017](https://doi.org/10.5281/zenodo.20878017). All plotted figures are rendered reproducibly from the raw logs by the `paper/build_*.py` scripts (`make figures`); the data figures in the main text are the laboratory’s actual recorded images, extracted verbatim. Alt text is provided under each figure and table caption per the AIP Publishing accessibility instructions.

SI Hardware



Figure S1: Fully disassembled graphite crucible. Top left is the crucible lid, top middle is the bottom of the crucible that acts as the specimen carrier, far right is the sapphire window, bottom left and middle are the alumina sample-surrounding sheets.

Alt text: Photograph of the disassembled graphite crucible parts: a stepped lid, the cup-shaped specimen carrier, a small transparent sapphire window disc, and two thin white alumina sheets.

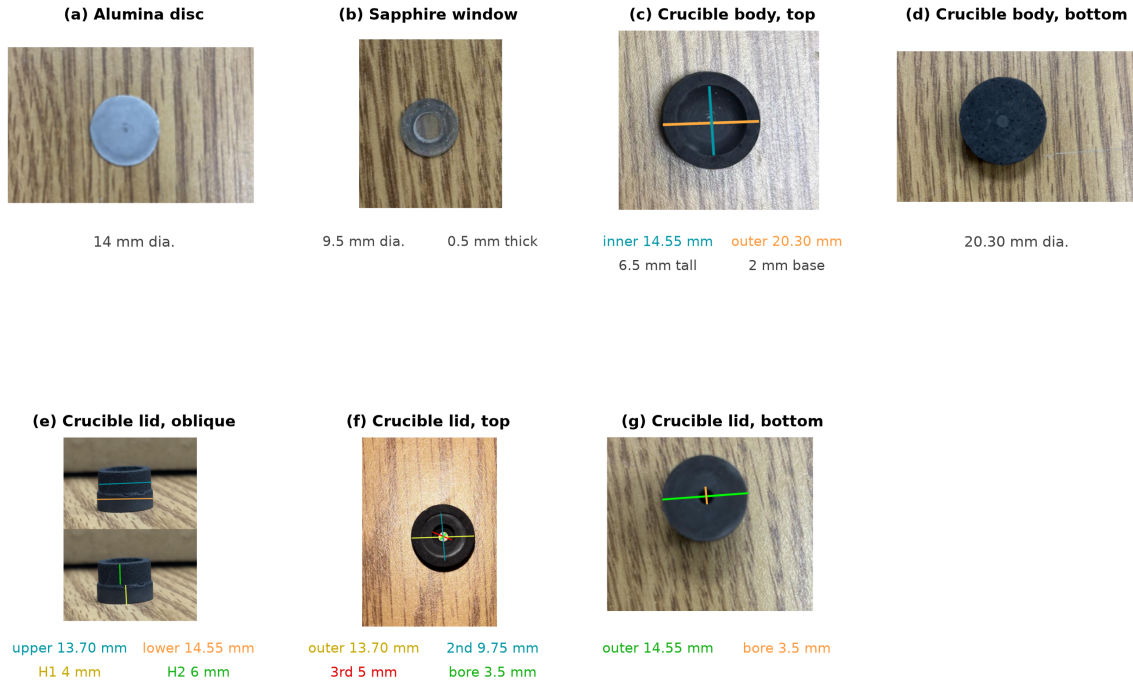


Figure S2: Measured dimensions of each part of the graphite crucible/susceptor stack, with the coloured measurement lines drawn on the corresponding photograph. (a) Alumina spacer disc, 14 mm dia. (b) Sapphire pyrometer window, 9.5 mm dia., 0.5 mm thick. (c) Crucible body, top: machined sample cavity 14.55 mm ID (teal), 20.30 mm OD (orange); the cup is 6.5 mm tall with a 2 mm base. (d) Crucible body, bottom face, 20.30 mm dia. (e) Crucible lid, oblique: the stepped plug seats into the cavity — 13.70 mm upper step (teal), 14.55 mm lower shoulder (orange); the upper step is 4 mm tall (H1, yellow) and the lower shoulder 6 mm tall (H2, green). (f) Crucible lid, top: four concentric steps of 13.70 mm (outermost), 9.75 mm, 5 mm, and a 3.5 mm central pyrometer-sighting bore. (g) Crucible lid, bottom: 14.55 mm seating face (green) and the 3.5 mm through-bore (orange). Fully assembled the stack stands 13 mm tall.

Alt text: Seven labeled photographs of the crucible parts, each overlaid with colored measurement lines and numeric callouts giving the diameters, step heights, and the central pyrometer bore dimensions.

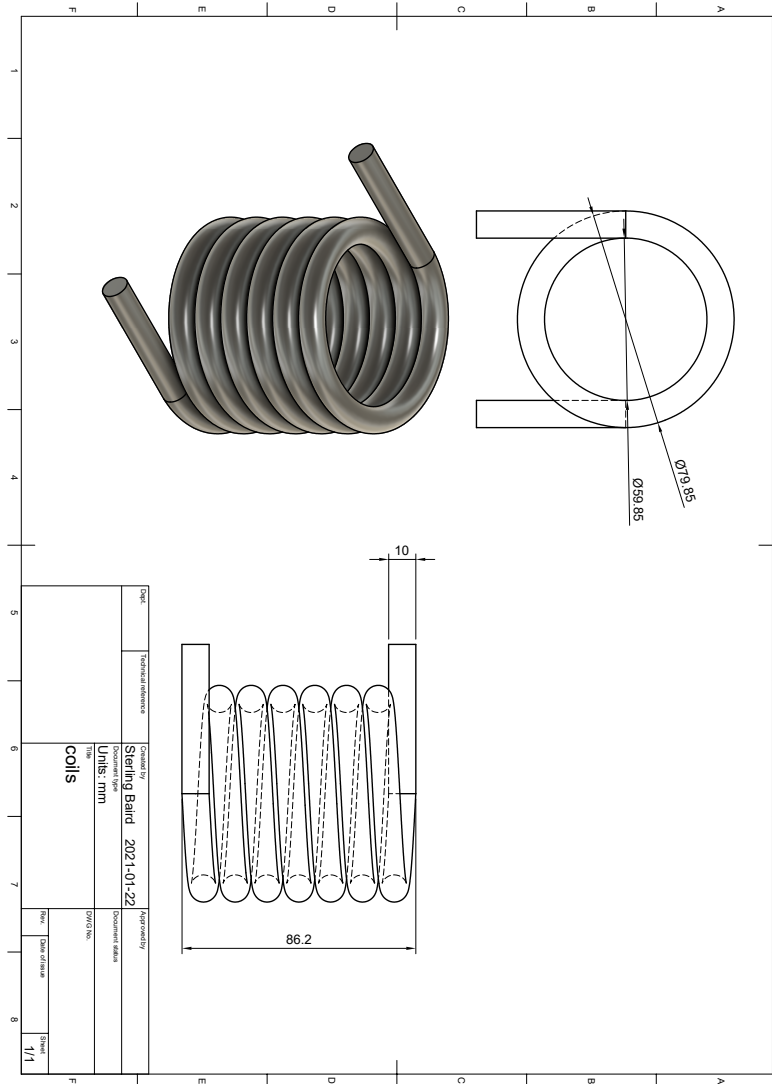
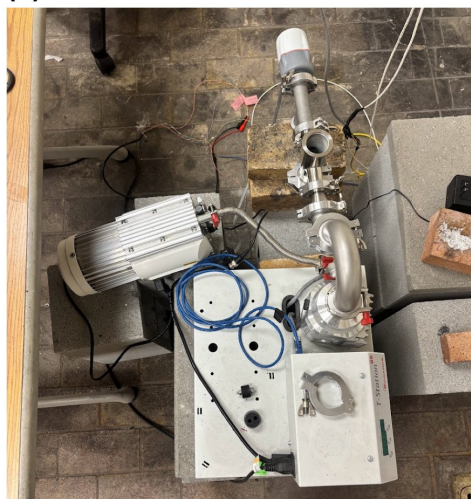


Figure S3: Work-coil fabrication drawing used for the current build, showing the coil geometry and dimensional callouts used to form and mount the liquid-cooled copper coil around the quartz chamber. (Source file: docs/coils-drawing.pdf.)

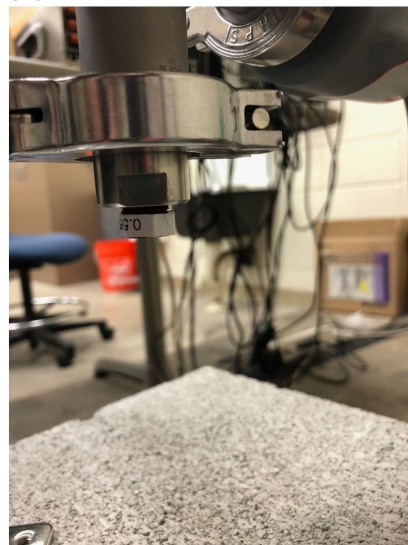
Alt text: Engineering drawing of the liquid-cooled copper work coil with dimensional callouts for the coil height, inner diameter, number of turns, and lead routing.

(a)



Pumping station: roughing pump on a separate support (vibration isolation)

(b)



Overpressure relief valve at the bottom of the chamber stack

Figure S4: Vacuum and gas-handling details. (a) Turbo pumping station with the roughing pump mounted on a separate support, coupled to the pumping station only through a flexible hose, so pump vibration is not transmitted into the chamber. (b) Overpressure relief valve (0.5 psi cracking pressure) at the bottom of the vacuum chamber stack.

Alt text: Two photographs: a turbo pumping station beside a separately supported roughing pump joined only by a flexible hose, and a small brass relief valve fitted at the base of the chamber stack.

SII Thermal performance

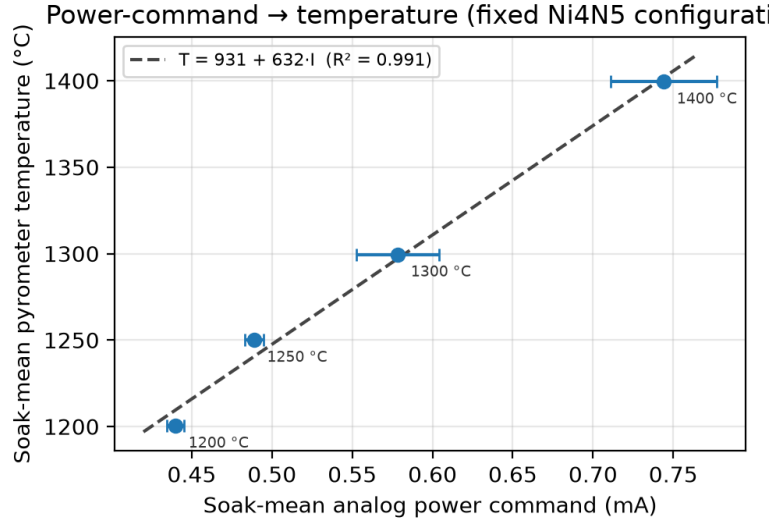


Figure S5: Power-command $\rightarrow$ temperature calibration for one fixed Ni4N5 configuration. Points are soak-mean values; horizontal bars are the in-soak standard deviation of the analog command. The dashed line is the linear operational calibration ($R^2 = 0.991$, $n = 4$).

Alt text: Scatter plot of soak-mean temperature against analog power command with four points between 1200 and 1400 degrees Celsius and a dashed straight fit line through them.

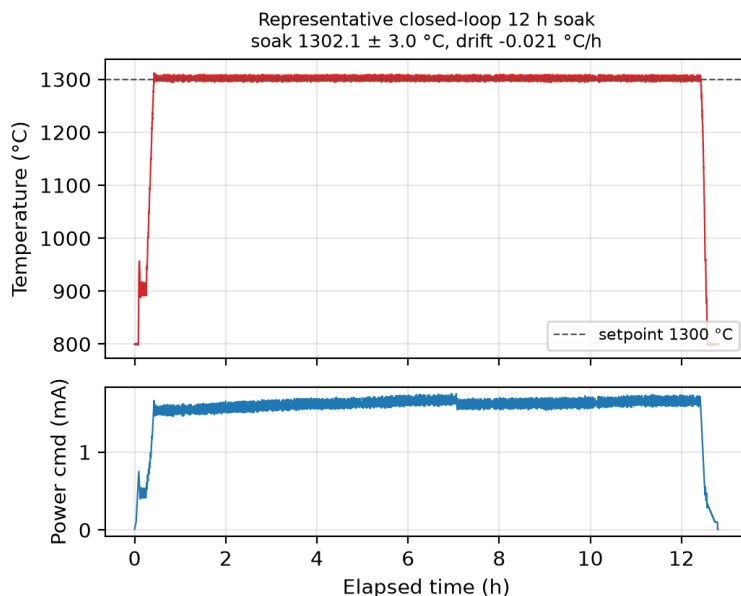


Figure S6: Representative closed-loop anneal (Ni₄N₅, nominal 1300 °C / 12 h): pyrometer temperature with setpoint overlay (top) and analog power command (bottom). Soak mean 1302.1 ± 3.0 °C, drift -0.02 °C/h.

Alt text: Two stacked line plots over twelve hours: the pyrometer temperature rising to and holding near 1300 degrees Celsius on its setpoint, and the analog power command settling to a steady value below.

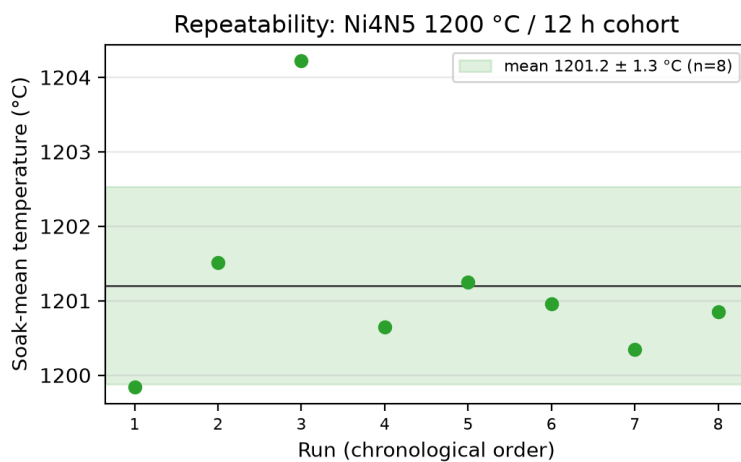


Figure S7: Repeatability of the Ni₄N₅ 1200 °C / 12 h cohort ($n = 8$). Points are per-run soak-mean temperatures; the band is mean $\pm$ SD (1201.2 ± 1.3 °C).

Alt text: Plot of eight per-run soak-mean temperatures clustered near 1200 degrees Celsius inside a shaded band marking the mean plus or minus one standard deviation.

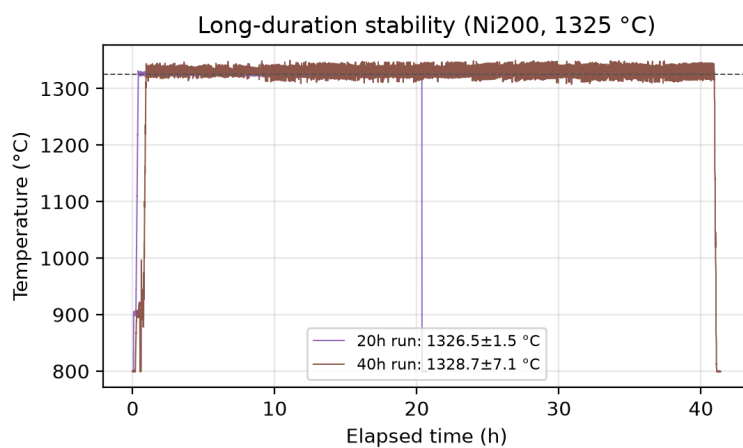


Figure S8: Long-duration thermal stability: Ni200 anneals at 1325 °C held for 20 h and 40 h.

Alt text: Line plot of two anneal temperature traces at 1325 degrees Celsius, one held twenty hours and one held forty hours, both essentially flat across the soak.

SIII Microstructure

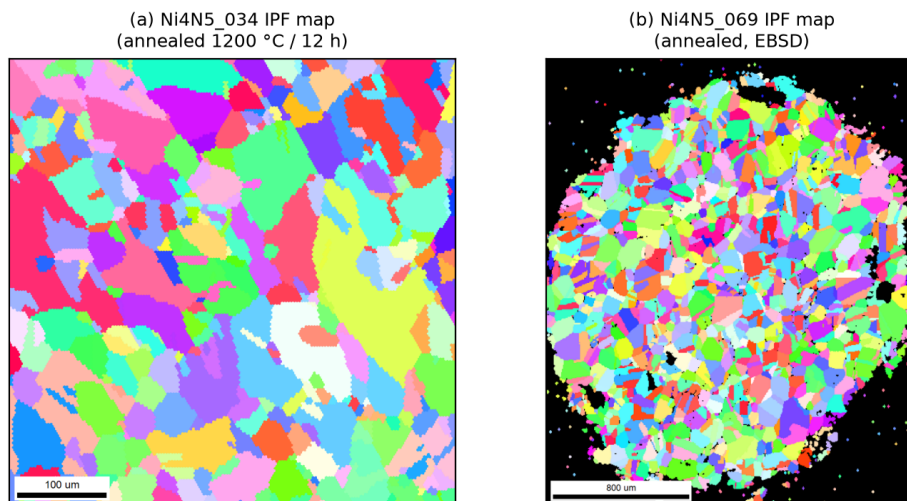


Figure S9: EBSD inverse-pole-figure (IPF) orientation maps of two annealed Ni4N5 specimens, rendered from the recorded EBSD scans. (a) Ni4N5_034 (annealed at 1200 °C / 12 h); (b) Ni4N5_069. Each indexed color is a crystallographic orientation; the maps resolve hundreds of grains for grain-size and texture analysis.

Alt text: Two colorful electron-backscatter-diffraction orientation maps in which each colored patch is a differently oriented grain, together resolving hundreds of grains.

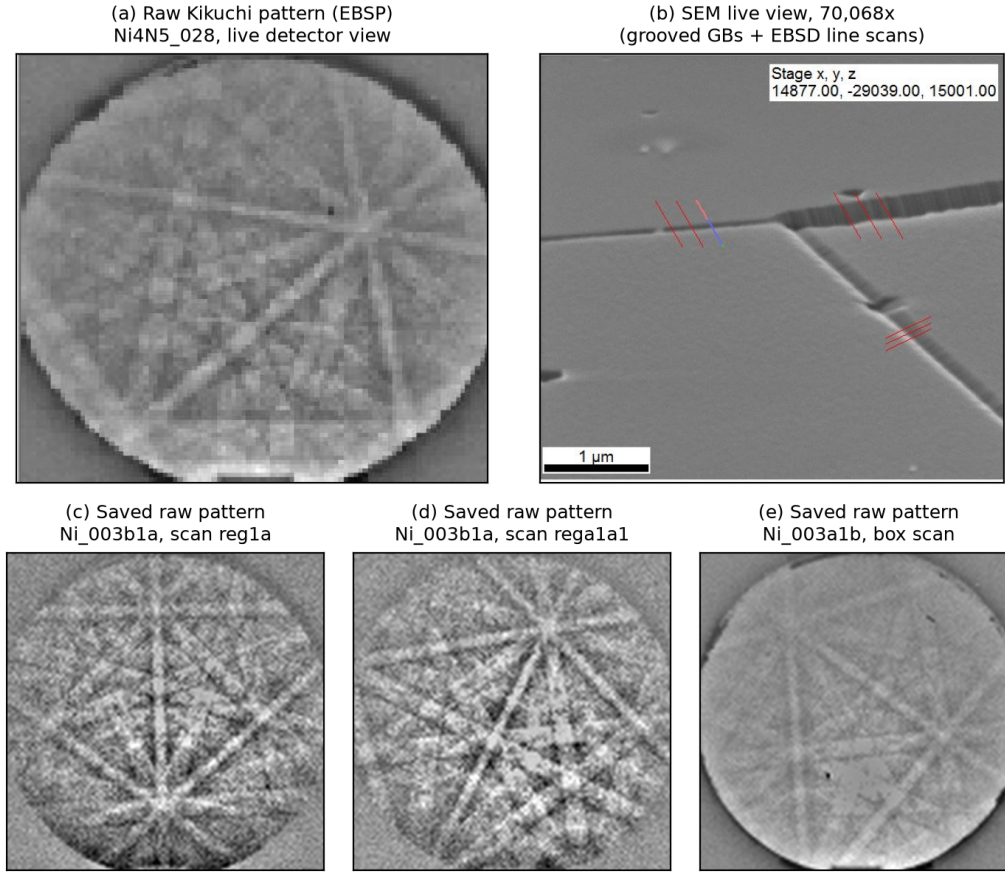


Figure S10: Raw (as-collected) EBSD acquisitions from furnace-annealed nickel, with no metallographic preparation. (a) Raw Kikuchi pattern (electron backscatter pattern) of specimen Ni4N5_028 as seen live on the detector at 8×8 binning (30 ms exposure), with Kikuchi bands and zone axes crisply resolved; (b) the corresponding SEM live view at 70,068 $\times$, where thermal grooving alone delineates the grain boundaries crossed by the EBSD line scans (red marks; 1 μm scale bar). (c–e) Representative per-scan-point raw patterns (115 $\times$ 115 px at 8×8 binning) saved automatically during EBSD mapping of the early-campaign specimens Ni_003b1a and Ni_003a1b; more than 100,000 such patterns are preserved in the repository’s raw-pattern archives ([docs/SEM/raw-kikuchi-patterns/](#)).

Alt text: Five grayscale panels: a raw Kikuchi diffraction pattern of intersecting bright bands, an electron micrograph with grooved grain boundaries and line-scan marks, and three small saved detector patterns.

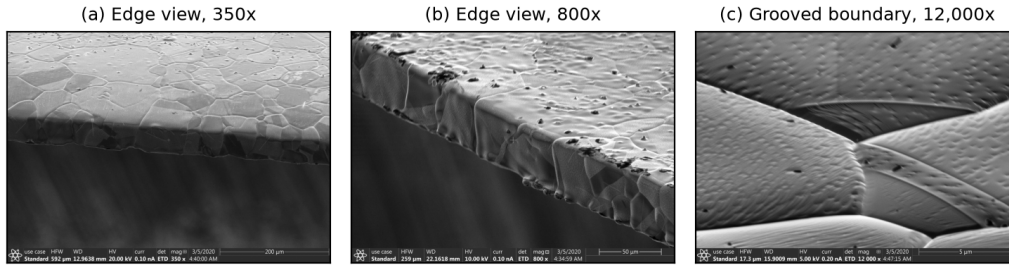


Figure S11: Grain-boundary thermal grooving on specimen Ni4N5.053 after a 1200 °C / 12 h anneal, imaged edge-on in the as-annealed condition (ultrasonic ethanol clean only; no polishing). (a) Edge view at 350 $\times$ (200 μ m scale bar): several grain boundaries traverse the full sheet thickness. (b) Edge and top surface at 800 $\times$ (50 μ m scale bar), with grooves outlining the surface grain structure. (c) A single grooved boundary at 12,000 $\times$ (5 μ m scale bar): the deep trench develops as grain-boundary energy equilibrates against surface energy at the soak temperature.

Alt text: Three electron micrographs at increasing magnification showing deep grooves along grain boundaries on the edge and top surface of an annealed nickel sheet.

SIV YSZ / high-temperature extension

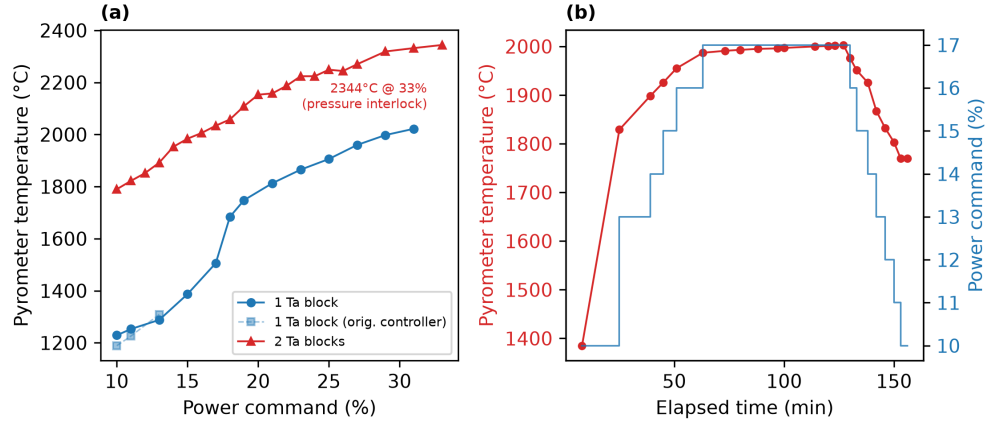


Figure S12: Measured heat curves of the tantalum-susceptor (ceramic) stack. (a) Pyrometer temperature vs. generator power command for one and two stacked tantalum susceptor blocks (the faint series repeats the single-block measurement on the original power controller); the two-block ramp reached 2344 °C at a 33% command before the chamber pressure interlock ended the test. (b) Timed two-block ramp/soak: a constant 17% command holds ~2000 °C (2000 ± 3 °C over the final hour) before a stepped ramp-down.

Alt text: Two plots: temperature versus power command curves for one and two tantalum blocks reaching 2344 degrees Celsius, and a timed ramp holding a steady 2000 degrees Celsius for over an hour.

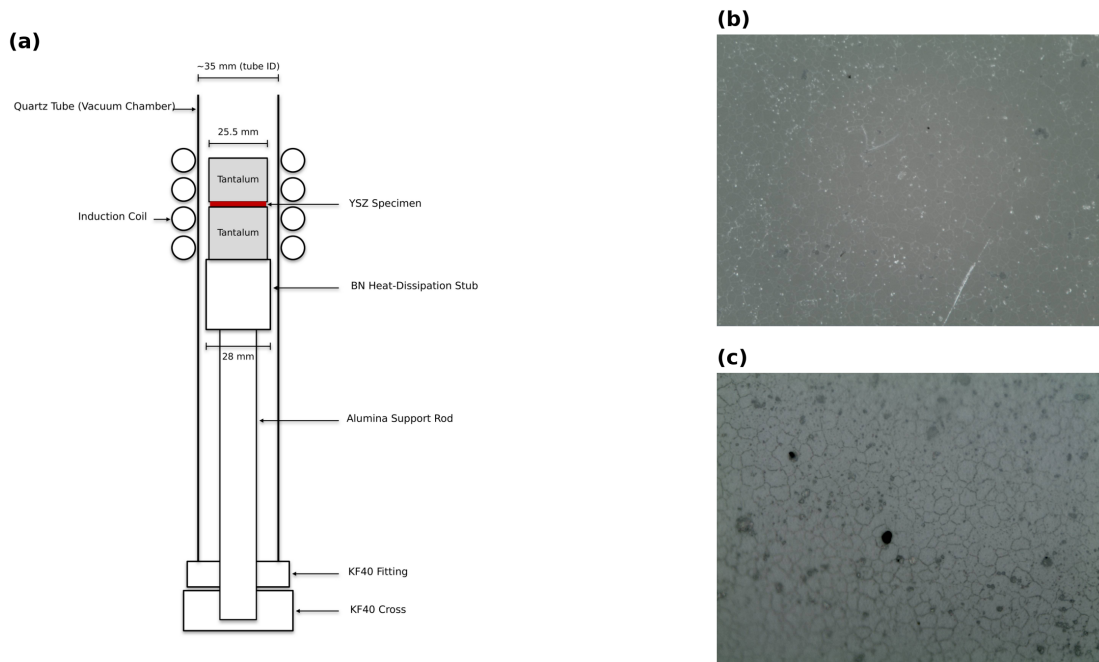


Figure S13: YSZ charge stack and resulting microstructure. (a) Charge-stack schematic: inside a fresh ~35 mm-ID quartz tube, the YSZ specimen (red) is sandwiched between two 25.5 mm tantalum susceptor blocks, resting on a 28 mm ceramic heat-dissipation stub (boron nitride for best results, sometimes MgO) on an alumina support rod that raises the stack to coil height; the coil, vacuum, pyrometer, and control paths are unchanged from the metal configuration. (b) Optical micrograph of YSZ after a 1700 °C / 10 h anneal. (c) Optical micrograph of an induction-annealed YSZ specimen, showing the equiaxed grain structure.

Alt text: A labeled schematic of the tantalum and YSZ charge stack inside a quartz tube, beside two optical micrographs of equiaxed YSZ grain structures.

SV Specimen–thermal-history linkage

Table [S1](#) links each characterized specimen to its furnace run and carries the run-ID-level detail deliberately kept out of the main text. Each run ID hyperlinks to the raw run log in the data repository. “Nominal soak” is the target parsed from the run filename; “Logged soak T” is the machine-parsed soak-mean from the trace (`docs/data_log/processed/run_summary.csv`); “Atmos.” indicates whether active gas flow was present in the parsed log summary, not a fully reconstructed gas chemistry; “Characterization” gives recorded SEM/optical file counts and flags recorded EBSD/OIM data. Machine-parsed per-run CSV traces and PNG plots for every linked run are in `docs/data_log/processed/`. The complete 35-specimen cross-reference, including characterized specimens whose run logs are not in the parsed set, is `paper/characterization-crossref.csv / .md`. Additional raw-pattern context for the Kikuchi figures (main text Fig. 4 and Fig. [S10](#)): the live-pattern screenshot of Ni4N5_028 is `docs/SEM/raw-kikuchi-patterns/200203_Ni4N5_028.s1/Scan3.JPG`, and the >100,000 preserved per-scan-point raw patterns of the early-campaign Ni_003-series specimens are inventoried in `docs/SEM/raw-kikuchi-patterns/README.md`.

Table S1: Specimen $\leftrightarrow$ thermal-history linkage for the runs used in the manuscript’s validation sections. Run IDs link to the raw run logs in the data repository.

Specimen	Material	Run (raw log)	Nominal soak	Logged soak T	Soak time	Atmos.	Characterization	Role / figure
Ni4N5_026	Ni4N5	IFrun039	1200 °C	1200.4 °C	6 h	no-flow	—	Calibration (Fig. S5)
Ni4N5_027	Ni4N5	IFrun040	1250 °C	1250.0 °C	6 h	no-flow	—	Calibration
Ni4N5_025	Ni4N5	IFrun038	1300 °C	1299.9 °C	1 h	no-flow	—	Calibration
Ni4N5_022	Ni4N5	IFrun032	1400 °C	1399.6 °C	10 min	no-flow	—	Calibration
Ni4N5_084	Ni4N5	IFrun079	1300 °C	1302.1 °C	12 h	flow	—	Representative soak (Fig. S6)
Ni4N5_040	Ni4N5	IFrun052	1200 °C	1199.8 °C	12 h	flow	—	Repeatability run 1 (Fig. S7)
Ni4N5_042,044	Ni4N5	IFrun054	1200 °C	1201.5 °C	12 h	flow	—	Repeatability run 2
Ni4N5_045,046	Ni4N5	IFrun055	1200 °C	1204.2 °C	12 h	flow	—	Repeatability run 3
Ni4N5_047,048	Ni4N5	IFrun056	1200 °C	1200.6 °C	12 h	flow	—	Repeatability run 4
Ni4N5_049,050	Ni4N5	IFrun057	1200 °C	1201.2 °C	12 h	flow	—	Repeatability run 5
Ni4N5_051,052	Ni4N5	IFrun058	1200 °C	1201.0 °C	12 h	flow	—	Repeatability run 6
Ni4N5_053,054	Ni4N5	IFrun059	1200 °C	1200.3 °C	12 h	flow	4 SEM, 3 opt. (Ni4N5_053)	Repeatability run 7; grooving (Fig. S11)
Ni4N5_056,057	Ni4N5	IFrun060	1200 °C	1200.9 °C	12 h	flow	—	Repeatability run 8
Ni200_015	Ni200	IFrun081	1325 °C	1326.5 °C	20 h	flow	3 opt.	Long soak, 20 h (Fig. S8)
Ni200_017	Ni200	IFrun080	1325 °C	1326.3 °C	40 h	flow	2 opt.	Long soak, 40 h (Fig. S8)
Ni4N5_079,080	Ni4N5	IFrun072	1400 °C	—	12 h	flow	—	Process-window specimen
Ni4N5_034	Ni4N5	IFrun049	1200 °C	1200.0 °C	12 h	no-flow	4 SEM, EBSD	EBSD IPF map (Fig. S9a)
Ni4N5_081	Ni4N5	IFrun082	1300 °C	1301.4 °C	20 h	flow	10 SEM, 10 opt., EBSD	Microstructure (main text Fig. 5)
Ni4N5_010	Ni4N5	IFrun016	1200 °C	1199.9 °C	—	no-flow	4 opt.	Characterization only
Ni4N5_012	Ni4N5	IFrun019	1200 °C	1199.6 °C	—	no-flow	1 opt.	Characterization only
Ni200_001	Ni200	IFrun004/IFrun005	—	—	—	no-flow	3 opt.	Characterization only
Ni4N5_001b	Ni4N5	IFrun006 (log not in parsed set)	—	—	—	no-flow	2 opt.	Characterization only

Alt text: Table linking each specimen to its furnace run, nominal and logged soak temperature, soak time, atmosphere, characterization file counts, and its role or figure in the manuscript.

SVI Bill of materials

Table S2 is the itemized bill of materials for the reproducible build; the main text’s Construction section cites the package-level cost and points here. Groupings by designator prefix: GEN = induction generator package (generator, controller, chiller, heating head, line transformer), RETRO = the reproducible retrofit (control / vacuum / sensing) parts, CONS = consumables. Full vendor/price detail for the RETRO and CONS parts is in `paper/extracted-context/parts_list.md` (extracted from `docs/induction_parts_list.xlsx`). All costs are in USD; prices are the laboratory’s purchase records (quote years 2019–2021), several from used/surplus sources as noted; shipping and tax are excluded unless stated. The GEN line items are from a single vendor quote (generator, heating head, controller, chiller, and line transformer totaling \$16,487). For the high-temperature ceramic (YSZ) configuration, the graphite crucible stack is exchanged for tantalum susceptor blocks resting on a ceramic heat-dissipation stub (boron nitride for best results, sometimes MgO); vendor records for the tantalum blocks and BN stubs are not in the archived parts list.

Table S2: Itemized bill of materials for the reproducible build (USD).

Desig.	Component	No.	Unit \$	Total	Source	Material
GEN-1	CEIA “Power Cube” PW3-90/50 solid-state RF generator (V3000-0070)	1	\$8,240	\$8,240	East Coast Induction (USA)	—
GEN-2	Power Controller C-V3 Plus (V3000-0414)	1	\$1,670	\$1,670	East Coast Induction (USA)	—
GEN-3	Recirculating chiller (MIL043008 air-cooled closed-loop, V1650-0060; vendor-listed as a water chiller, but the loop is charged with ethylene glycol coolant)	1	\$1,575	\$1,575	East Coast Induction (USA)	—
GEN-4	Heating head + 3 m cable and coolant lines (PWH-13-12-30/50, V3000-0300); 1 coil made to spec at no charge with a complete unit	1	\$3,337	\$3,337	East Coast Induction (USA)	copper
GEN-5	Line auto-transformer, 12 kVA / 3PH / 480×380 V with taps (V049-0508; facility-dependent)	1	\$1,665	\$1,665	East Coast Induction (USA)	—
RETRO-1	LabVIEW-compatible DAQ with analog out + in (0–5 V AO): NI USB-6000 DAQ	1	\$281.00	\$281.00	National Instruments	—
RETRO-2	Dual-wavelength ratio pyrometer (LumaSense IMPAC ISR 6, 800–2500 °C), used	1	\$385.00 (used)	\$385.00	eBay — used listing (new list ~\$5,500, LumaSense quote 00161403)	—
RETRO-3	24 V linear power supply for pyrometer (International Power)	1	\$51	\$51	Mouser	—
RETRO-4	Voltage-to-current (0–5 V → 4–20 mA) loop conditioner set for generator power-setpoint input (NI-9265 input module + NI-9203 output module + NI cDAQ-9174 chassis)	1 set	\$3,755.00	\$3,755.00	National Instruments	—
RETRO-5	Edwards nEXT T-Station 85H Dry turbo pumping station (1Ph 100–120 V, 50/60 Hz, DN 40 ISO-KF)	1	\$13,573.65	\$13,573.65	Edwards Vacuum	—
RETRO-5A	Wide-range vacuum gauge D14701000 (36 V, 2 W); discontinued, replaced by D3G0021100	1	\$1,543.29	\$1,543.29	Chemtech Scientific	—
RETRO-6	Edwards TAV5 vent valve (new)	1	\$652.80	\$652.80	Edwards Vacuum	—
RETRO-6A	Sierra SmartTrak 100L mass flow controller (optional; helpful when continuous inert gas flow is required)	1	\$550–\$1,000 (quote range)	\$550–\$1,000	Sierra / distributor quote	—
RETRO-7	Inert-gas regulator (Fisher FS-50)	1	\$38	\$38	Fisher Scientific	—
RETRO-8	KF40 overpressure centering ring	1	\$19	\$19	IdealVac	aluminum
RETRO-9	KF40 plastic quick vacuum clamp	1	\$23	\$23	IdealVac	polymer
RETRO-10	Optical window, custom quartz disc 55 mm × 1.5 mm (1 used; buy 3 for spares)	3	\$18 (est.)	\$54	Custom-cut metric disc — vendor not recorded in parts list; price approximate (McMaster stocks only imperial sizes)	fused quartz
RETRO-11	Ultra-high-temperature quartz disc 2 in × 1/16 in (McMaster 1357T12)	1	\$19	\$19	McMaster-Carr	fused quartz
RETRO-12	Inert-gas plumbing (BSPT/KF adapters, barbs, tee, 10 psi relief valve McMaster 4772K4)	1 set	see parts list	~\$66	McMaster-Carr / BMotionTech	brass / steel
CONS-1	Quartz tube, 4 ft × 35 mm ID, cut to length	2	\$67	\$134	QSI Quartz	fused quartz
CONS-2	Graphite stock for machined crucible/susceptor (56L-3, 3000 °C, 1×6×6 in)	1	\$99	\$99	Cotronics	graphite
CONS-3	Graphite repair cement (Resbond 931-1, 3000 °C)	1	\$108	\$108	Cotronics	graphite
CONS-4	Alumina crucible stock (RTC-60-2, 1787 °C, 10 lb)	1	\$91	\$91	Cotronics	alumina
CONS-5	Zirconia crucible stock (760-1, 2204 °C, 10 lb)	1	\$124	\$124	Cotronics	zirconia
CONS-6	Torr-Seal high-vacuum epoxy / O-rings	1 set	\$57	\$57	Varian / McMaster	epoxy / elastomer

Alt text: Itemized bill of materials with designator, component description, quantity, unit and total cost in US dollars, source, and material for each generator, retrofit, and consumable part.

SVII Design-file inventory

Table S3 lists the open design files in the repository (paths relative to the repository root; the archival snapshot is the Zenodo deposit above). The graphite crucible/susceptor machining drawing is pending; its measured dimensions are given in Fig. S2. Exports of the LabVIEW block diagrams to PDF/PNG for non-LabVIEW readers are likewise pending.

Table S3: Open design files in the repository.

Design file	File type	License	Location
LabVIEW control VIs (furnace control, manual control, PID tuning v1/v2, email alert)	LabVIEW .vi	MIT (proposed)	<code>code/induction-furnace-control-code/</code>
Manual ramping v5 VIs	LabVIEW .vi	MIT (proposed)	<code>code/manual-ramping-v5/</code>
<code>plotheatcurve.m</code> (heat-curve plotter)	MATLAB	MIT (proposed)	<code>code/plotheatcurve.m</code>
Work-coil drawing	PDF / OXPS	CERN-OHL-S (proposed)	<code>docs/coils-drawing.pdf</code> , <code>docs/coils.oxps</code>
System schematics	PPTX (+ extracted text/figures)	CERN-OHL-S (proposed)	<code>docs/*.pptx</code> , <code>paper/extracted-context/</code>
Temperature-control wiring	PNG	CERN-OHL-S (proposed)	<code>docs/temp-control-modification/</code>
KF40 overpressure centering ring	STEP	CERN-OHL-S (proposed)	<code>docs/KF Supplies/</code>
YSZ / tantalum-susceptor stack: schematic, heat-curve workbook, configuration survey, compatibility notes	PNG / XLSX / PDF / DOCX	CERN-OHL-S (proposed)	<code>docs/YSZ/</code>
Graphite crucible/susceptor machining drawing	pending	CERN-OHL-S (proposed)	pending (dimensions in Fig. S2)

Alt text: Table listing each open design file, its file type, proposed license, and its location in the repository.